Chapter 1

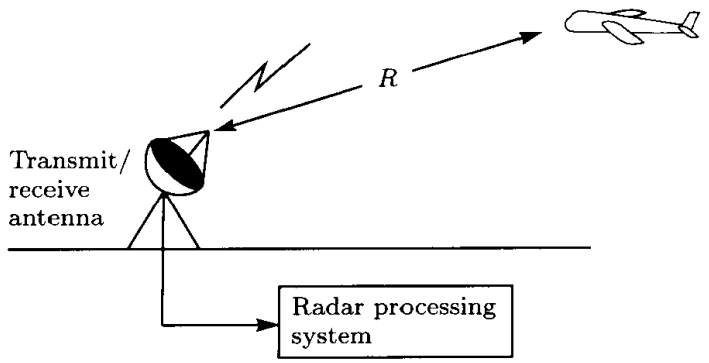
Introduction

1.1 Estimation in Signal Processing

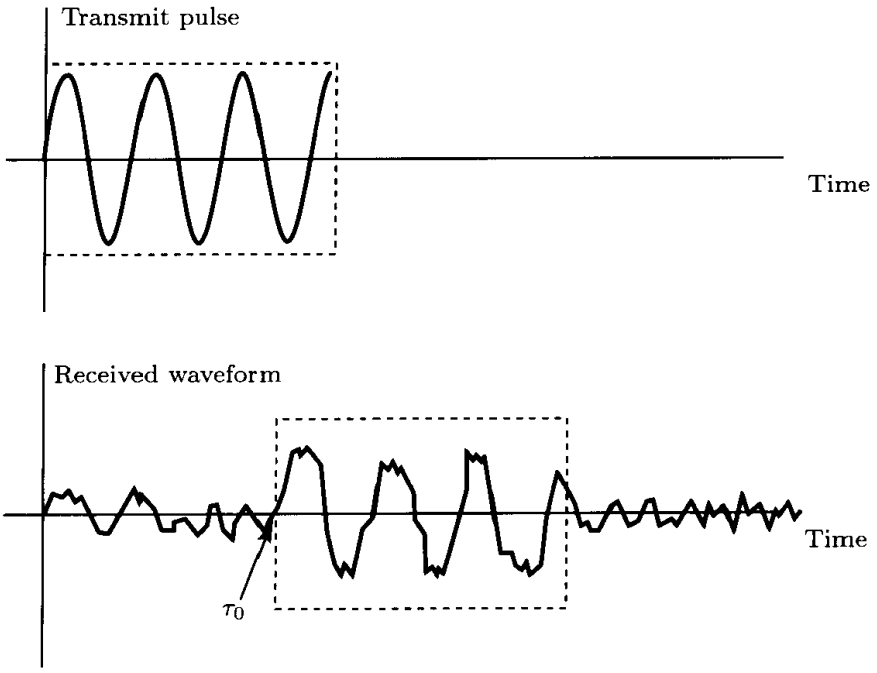
Modern estimation theory can be found at the heart of many electronic signal processing systems designed to extract information. These systems include

1. Radar
2. Sonar
3. Speech
4. Image analysis
5. Biomedicine
6. Communications
7. Control
8. Seismology,

and all share the common problem of needing to estimate the values of a group of parameters. We briefly describe the first three of these systems. In radar we are interested in determining the position of an aircraft, as for example, in airport surveillance radar [Skolnik 1980]. To determine the range R we transmit an electromagnetic pulse that is reflected by the aircraft, causing an echo to be received by the antenna τ_0 seconds later, as shown in Figure 1.1a. The range is determined by the equation $\tau_0 = 2R/c$, where c is the speed of electromagnetic propagation. Clearly, if the round trip delay τ_0 can be measured, then so can the range. A typical transmit pulse and received waveform are shown in Figure 1.1b. The received echo is decreased in amplitude due to propagation losses and hence may be obscured by environmental noise. Its onset may also be perturbed by time delays introduced by the electronics of the receiver. Determination of the round trip delay can therefore require more than just a means of detecting a jump in the power level at the receiver. It is important to note that a typical modern



(a) Radar

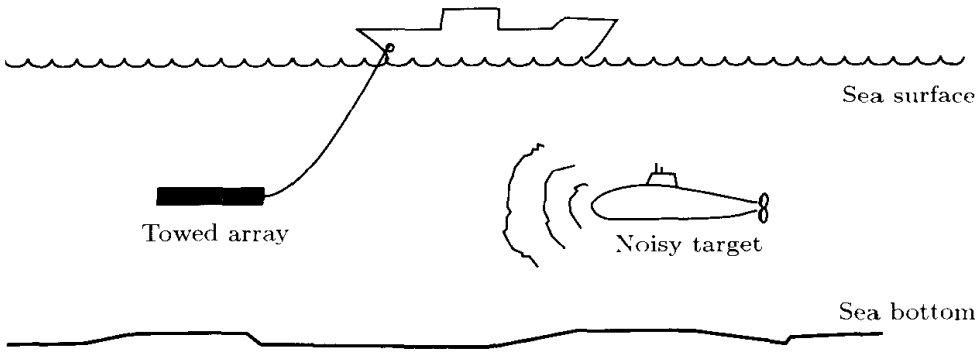


(b) Transmit and received waveforms

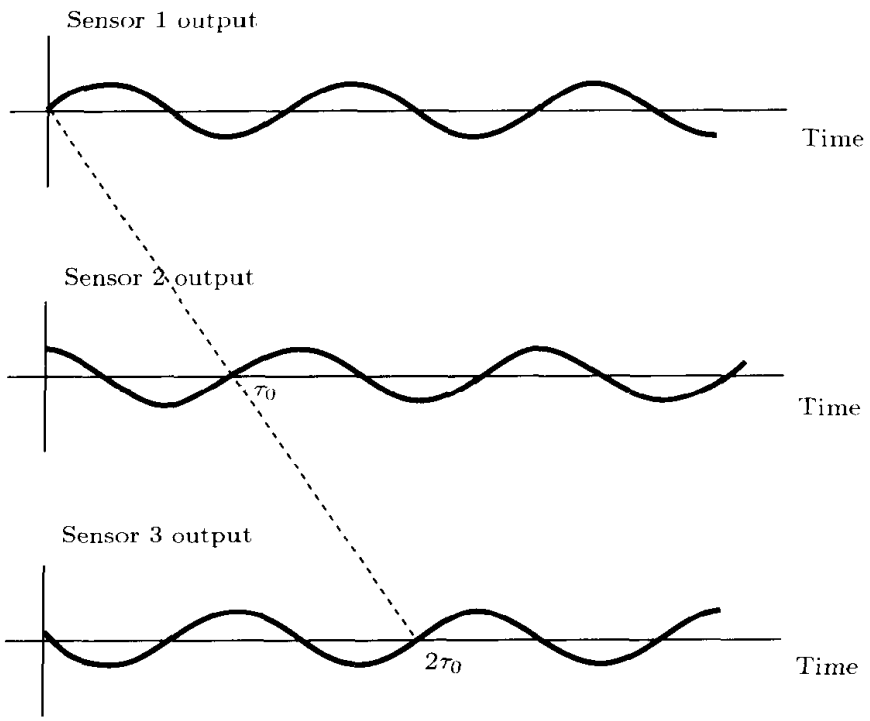
Figure 1.1 Radar system

radar system will input the received continuous-time waveform into a digital computer by taking samples via an analog-to-digital convertor. Once the waveform has been sampled, the data compose a *time series.* (See also Examples 3.13 and 7.15 for a more detailed description of this problem and optimal estimation procedures.)

Another common application is in sonar, in which we are also interested in the position of a target, such as a submarine [Knight et al. 1981, Burdic 1984] . A typical passive sonar is shown in Figure 1.2a. The target radiates noise due to machinery on board, propellor action, etc. This noise, which is actually the *signal* of interest, propagates through the water and is received by an array of sensors. The sensor outputs



(a) Passive sonar



(b) Received signals at array sensors

Figure 1.2 Passive sonar system

are then transmitted to a tow ship for input to a digital computer. Because of the positions of the sensors relative to the arrival angle of the target signal, we receive the signals shown in Figure 1.2b. By measuring τ_0, the delay between sensors, we can determine the bearing β from the expression

$$\beta = \arccos\left(\frac{c\tau_0}{d}\right) \tag{1.1}$$

where c is the speed of sound in water and d is the distance between sensors (see Examples 3.15 and 7.17 for a more detailed description). Again, however, the received

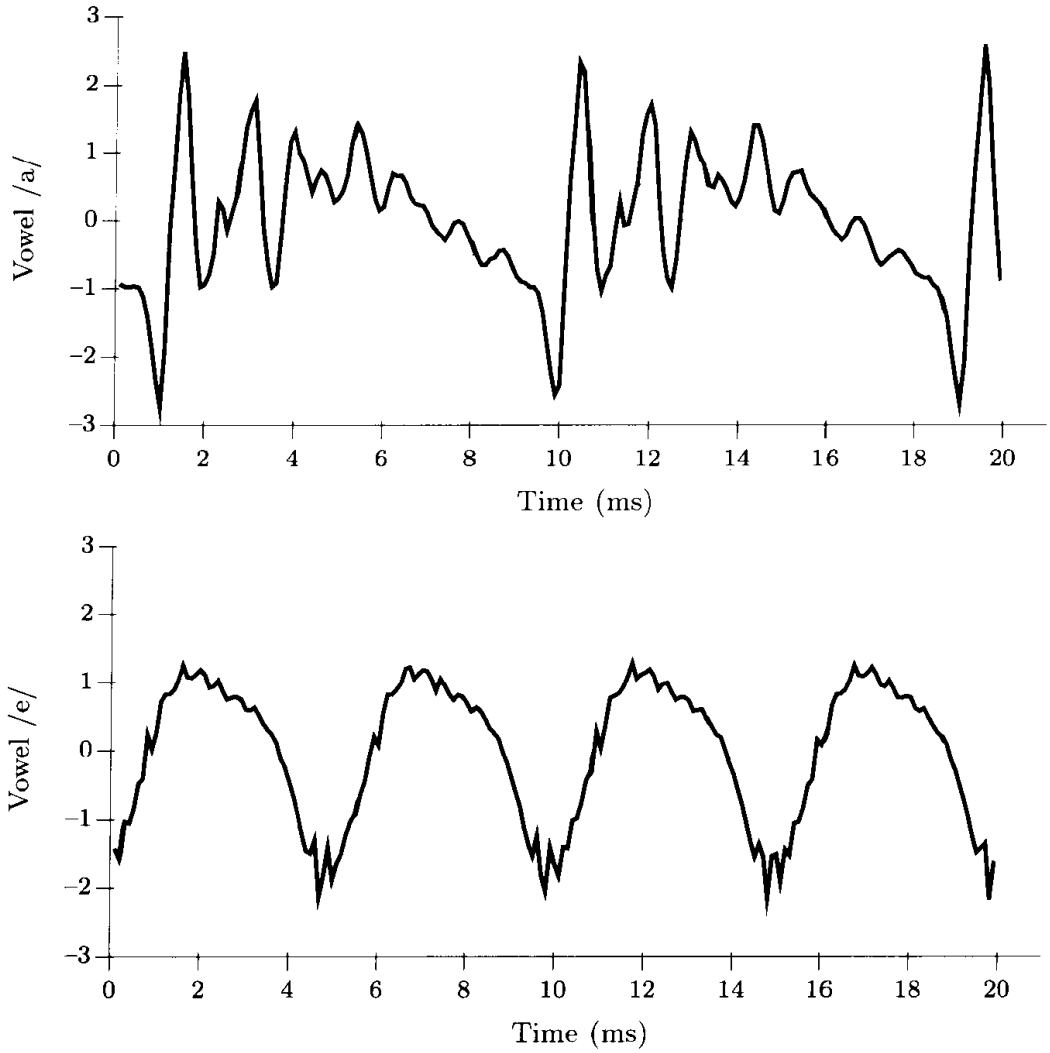


Figure 1.3 Examples of speech sounds

waveforms are not "clean" as shown in Figure 1.2b but are embedded in noise, making the determination of τ_0 more difficult. The value of β obtained from (1.1) is then only an estimate.

Another application is in speech processing systems [Rabiner and Schafer 1978]. A particularly important problem is speech recognition, which is the recognition of speech by a machine (digital computer). The simplest example of this is in recognizing individual speech sounds or *phonemes*. Phonemes are the vowels, consonants, etc., or the fundamental sounds of speech. As an example, the vowels /a/ and /e/ are shown in Figure 1.3. Note that they are periodic waveforms whose period is called the *pitch*. To recognize whether a sound is an /a/ or an /e/ the following simple strategy might be employed. Have the person whose speech is to be recognized say each vowel three times and store the waveforms. To recognize the spoken vowel, compare it to the stored vowels and choose the one that is closest to the spoken vowel or the one that

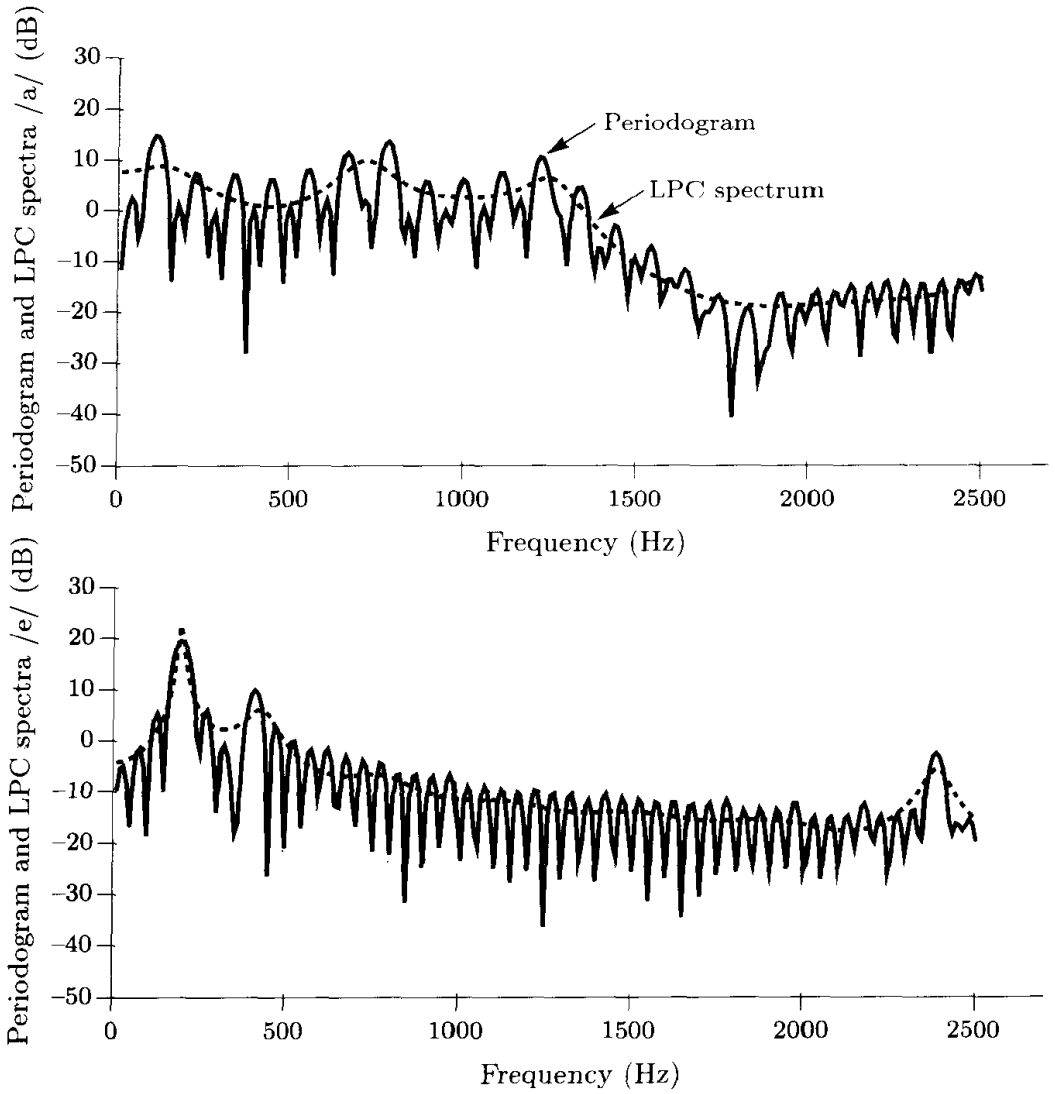


Figure 1.4 LPC spectral modeling

minimizes some distance measure. Difficulties arise if the pitch of the speaker's voice changes from the time he or she records the sounds (the training session) to the time when the speech recognizer is used. This is a natural variability due to the nature of human speech. In practice, attributes, other than the waveforms themselves, are used to measure distance. Attributes are chosen that are less susceptible to variation. For example, the spectral envelope will not change with pitch since the Fourier transform of a periodic signal is a sampled version of the Fourier transform of one period of the signal. The period affects only the spacing between frequency samples, not the values. To extract the spectral envelope we employ a model of speech called *linear predictive coding* (LPC). The parameters of the model determine the spectral envelope. For the speech sounds in Figure 1.3 the power spectrum (magnitude-squared Fourier transform divided by the number of time samples) or periodogram and the estimated LPC spectral envelope are shown in Figure 1.4. (See Examples 3.16 and 7.18 for a description of how

the parameters of the model are estimated and used to find the spectral envelope.) It is interesting that in this example a human interpreter can easily discern the spoken vowel. The real problem then is to design a machine that is able to do the same. In the radar/sonar problem a human interpreter would be unable to determine the target position from the received waveforms, so that the machine acts as an indispensable tool.

In all these systems we are faced with the problem of extracting values of parameters based on continuous-time waveforms. Due to the use of digital computers to sample and store the continuous-time waveform, we have the equivalent problem of extracting parameter values from a *discrete-time* waveform or a *data* set. Mathematically, we have the N-point data set $\{x[0], x[1], \ldots, x[N-1]\}$ which depends on an unknown parameter θ. We wish to determine θ based on the data or to define an *estimator*

$$\hat{\theta} = g(x[0], x[1], \ldots, x[N-1]) \tag{1.2}$$

where g is some function. This is the problem of *parameter estimation*, which is the subject of this book. Although electrical engineers at one time designed systems based on analog signals and analog circuits, the current and future trend is based on discrete-time signals or sequences and digital circuitry. With this transition the estimation problem has evolved into one of estimating a parameter based on a *time series*, which is just a discrete-time process. Furthermore, because the amount of data is necessarily finite, we are faced with the determination of g as in (1.2). Therefore, our problem has now evolved into one which has a long and glorious history, dating back to Gauss who in 1795 used least squares data analysis to predict planetary movements [Gauss 1963 (English translation)]. All the theory and techniques of statistical estimation are at our disposal [Cox and Hinkley 1974, Kendall and Stuart 1976–1979, Rao 1973, Zacks 1981].

Before concluding our discussion of application areas we complete the previous list.

4. Image analysis - estimate the position and orientation of an object from a camera image, necessary when using a robot to pick up an object [Jain 1989]

5. Biomedicine - estimate the heart rate of a fetus [Widrow and Stearns 1985]

6. Communications - estimate the carrier frequency of a signal so that the signal can be demodulated to baseband [Proakis 1983]

7. Control - estimate the position of a powerboat so that corrective navigational action can be taken, as in a LORAN system [Dabbous 1988]

8. Seismology - estimate the underground distance of an oil deposit based on sound reflections due to the different densities of oil and rock layers [Justice 1985].

Finally, the multitude of applications stemming from analysis of data from physical experiments, economics, etc., should also be mentioned [Box and Jenkins 1970, Holm and Hovem 1979, Schuster 1898, Taylor 1986].

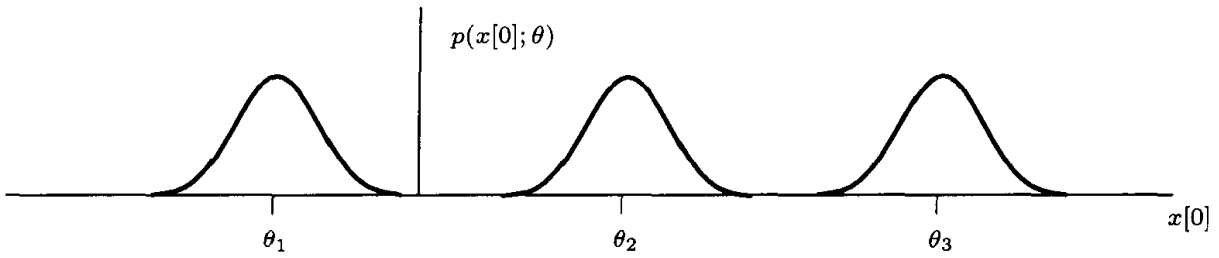


Figure 1.5 Dependence of PDF on unknown parameter

1.2 The Mathematical Estimation Problem

In determining good estimators the first step is to mathematically model the data. Because the data are inherently random, we describe it by its probability density function (PDF) or $p(x[0], x[1], \ldots, x[N-1]; \theta)$. The PDF is *parameterized* by the unknown parameter θ, i.e., we have a class of PDFs where each one is different due to a different value of θ. We will use a semicolon to denote this dependence. As an example, if $N = 1$ and θ denotes the mean, then the PDF of the data might be

$$p(x[0];\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}(x[0]-\theta)^2\right]$$

which is shown in Figure 1.5 for various values of θ. It should be intuitively clear that because the value of θ affects the probability of $x[0]$, we should be able to *infer* the value of θ from the observed value of $x[0]$. For example, if the value of $x[0]$ is negative, it is doubtful that $\theta = \theta_2$. The value $\theta = \theta_1$ might be more reasonable. This specification of the PDF is critical in determining a good estimator. In an actual problem we are not given a PDF but must choose one that is not only consistent with the problem constraints and any prior knowledge, but one that is also mathematically tractable. To illustrate the approach consider the hypothetical Dow-Jones industrial average shown in Figure 1.6. It might be conjectured that this data, although appearing to fluctuate wildly, actually is "on the average" increasing. To determine if this is true we could assume that the data actually consist of a straight line embedded in random noise or

$$x[n] = A + Bn + w[n] \qquad n = 0, 1, \ldots, N-1.$$

A reasonable model for the noise is that $w[n]$ is white Gaussian noise (WGN) or each sample of $w[n]$ has the PDF $\mathcal{N}(0, \sigma^2)$ (denotes a Gaussian distribution with a mean of 0 and a variance of σ^2) and is uncorrelated with all the other samples. Then, the unknown parameters are A and B, which arranged as a vector become the vector parameter $\boldsymbol{\theta} = [A\, B]^T$. Letting $\mathbf{x} = [x[0]\, x[1] \ldots x[N-1]]^T$, the PDF is

$$p(\mathbf{x};\boldsymbol{\theta}) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A-Bn)^2\right]. \tag{1.3}$$

The choice of a straight line for the signal component is consistent with the knowledge that the Dow-Jones average is hovering around 3000 (A models this) and the conjecture

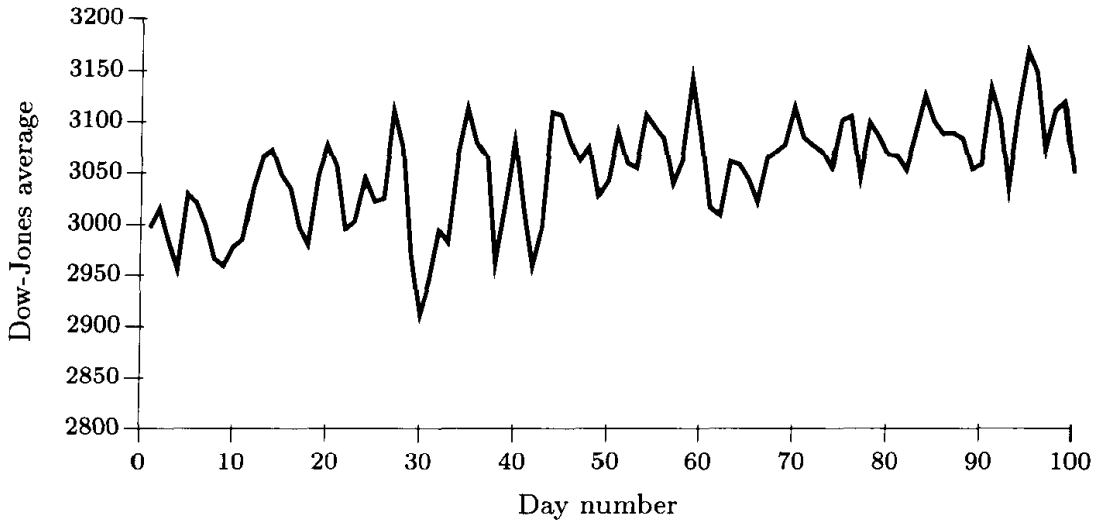


Figure 1.6 Hypothetical Dow-Jones average

that it is increasing ($B > 0$ models this). The assumption of WGN is justified by the need to formulate a mathematically tractable model so that closed form estimators can be found. Also, it is reasonable unless there is strong evidence to the contrary, such as highly correlated noise. Of course, the performance of any estimator obtained will be *critically dependent* on the PDF assumptions. We can only hope the estimator obtained is robust, in that slight changes in the PDF do not severely affect the performance of the estimator. More conservative approaches utilize robust statistical procedures [Huber 1981].

Estimation based on PDFs such as (1.3) is termed *classical* estimation in that the parameters of interest are assumed to be *deterministic* but unknown. In the Dow-Jones average example we know a priori that the mean is somewhere around 3000. It seems inconsistent with reality, then, to choose an estimator of A that can result in values as low as 2000 or as high as 4000. We might be more willing to constrain the estimator to produce values of A in the range [2800, 3200]. To incorporate this prior knowledge we can assume that A is no longer deterministic but a random variable and assign it a PDF, possibly uniform over the [2800, 3200] interval. Then, any subsequent estimator will yield values in this range. Such an approach is termed *Bayesian* estimation. The parameter we are attempting to estimate is then viewed as a *realization* of the random variable θ. As such, the data are described by the *joint* PDF

$$p(\mathbf{x}, \theta) = p(\mathbf{x}|\theta)p(\theta)$$

where $p(\theta)$ is the prior PDF, summarizing our knowledge about θ before any data are observed, and $p(\mathbf{x}|\theta)$ is a conditional PDF, summarizing our knowledge provided by the data $\mathbf{x}$ conditioned on knowing θ. The reader should compare the notational differences between $p(\mathbf{x}; \theta)$ (a family of PDFs) and $p(\mathbf{x}|\theta)$ (a conditional PDF), as well as the implied interpretations (see also Problem 1.3).

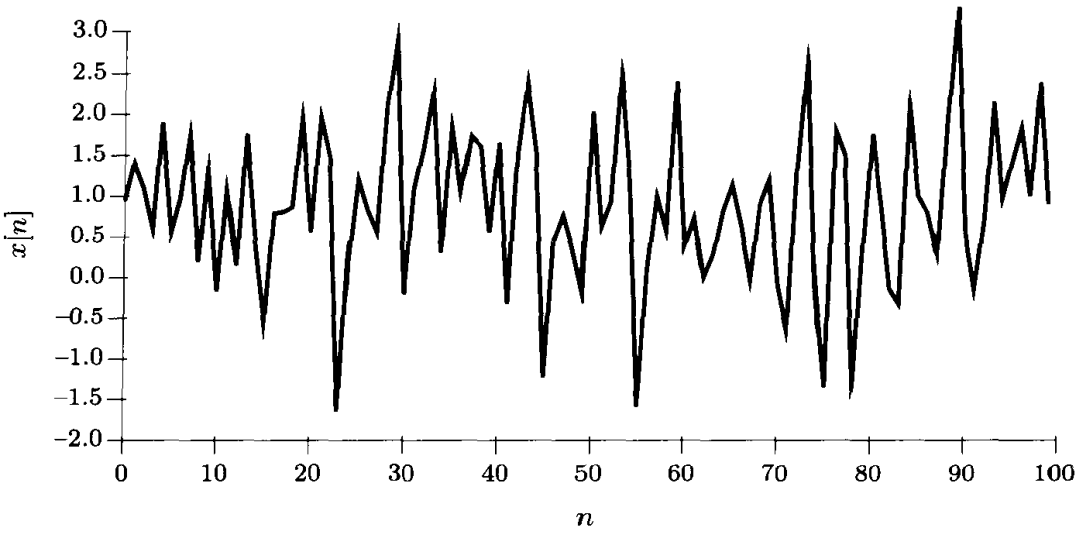


Figure 1.7 Realization of DC level in noise

Once the PDF has been specified, the problem becomes one of determining an optimal estimator or function of the data, as in (1.2). Note that an estimator may depend on other parameters, but only if they are *known*. An *estimator* may be thought of as a *rule* that assigns a value to θ for each realization of $\mathbf{x}$. The *estimate* of θ is the *value* of θ obtained for a *given* realization of $\mathbf{x}$. This distinction is analogous to a random variable (which is a function defined on the sample space) and the value it takes on. Although some authors distinguish between the two by using capital and lowercase letters, we will not do so. The meaning will, hopefully, be clear from the context.

1.3 Assessing Estimator Performance

Consider the data set shown in Figure 1.7. From a cursory inspection it appears that $x[n]$ consists of a DC level A in noise. (The use of the term DC is in reference to direct current, which is equivalent to the constant function.) We could model the data as

$$x[n] = A + w[n]$$

where $w[n]$ denotes some zero mean noise process. Based on the data set $\{x[0], x[1], \ldots, x[N-1]\}$, we would like to estimate A. Intuitively, since A is the average level of $x[n]$ ($w[n]$ is zero mean), it would be reasonable to estimate A as

$$\hat{A} = \frac{1}{N} \sum_{n=0}^{N-1} x[n]$$

or by the sample mean of the data. Several questions come to mind:

1. How close will $\hat{A}$ be to A?

2. Are there better estimators than the sample mean?

For the data set in Figure 1.7 it turns out that $\hat{A} = 0.9$, which is close to the true value of $A = 1$. Another estimator might be

$$\check{A} = x[0].$$

Intuitively, we would not expect this estimator to perform as well since it does not make use of all the data. There is no *averaging* to reduce the noise effects. However, for the data set in Figure 1.7, $\check{A} = 0.95$, which is closer to the true value of A than the sample mean estimate. Can we conclude that $\check{A}$ is a better estimator than $\hat{A}$? The answer is of course no. Because an estimator is a function of the data, which are random variables, it too is a random variable, subject to many possible outcomes. The fact that $\check{A}$ is closer to the true value only means that for the *given realization of data*, as shown in Figure 1.7, the estimate $\check{A} = 0.95$ (or realization of $\check{A}$) is closer to the true value than the estimate $\hat{A} = 0.9$ (or realization of $\hat{A}$). To assess performance we must do so statistically. One possibility would be to repeat the experiment that generated the data and apply each estimator to every data set. Then, we could ask which estimator produces a better estimate in the majority of the cases. Suppose we repeat the experiment by fixing $A = 1$ and adding different noise realizations of $w[n]$ to generate an ensemble of realizations of $x[n]$. Then, we determine the values of the two estimators for each data set and finally plot the histograms. (A histogram describes the number of times the estimator produces a given range of values and is an approximation to the PDF.) For 100 realizations the histograms are shown in Figure 1.8. It should now be evident that $\hat{A}$ is a better estimator than $\check{A}$ because the values obtained are more concentrated about the true value of $A = 1$. Hence, $\hat{A}$ will *usually* produce a value closer to the true one than $\check{A}$. The skeptic, however, might argue that if we repeat the experiment 1000 times instead, then the histogram of $\check{A}$ will be more concentrated. To dispel this notion, we cannot repeat the experiment 1000 times, for surely the skeptic would then reassert his or her conjecture for 10,000 experiments. To *prove* that $\hat{A}$ is better we could establish that the variance is less. The modeling assumptions that we must employ are that the $w[n]$'s, in addition to being zero mean, are uncorrelated and have equal variance σ^2. Then, we first show that the mean of each estimator is the true value or

$$\begin{aligned} E(\hat{A}) &= E\left(\frac{1}{N}\sum_{n=0}^{N-1} x[n]\right) \\ &= \frac{1}{N}\sum_{n=0}^{N-1} E(x[n]) \\ &= A \\ E(\check{A}) &= E(x[0]) \\ &= A \end{aligned}$$

so that on the average the estimators produce the true value. Second, the variances are

$$\text{var}(\hat{A}) = \text{var}\left(\frac{1}{N}\sum_{n=0}^{N-1} x[n]\right)$$

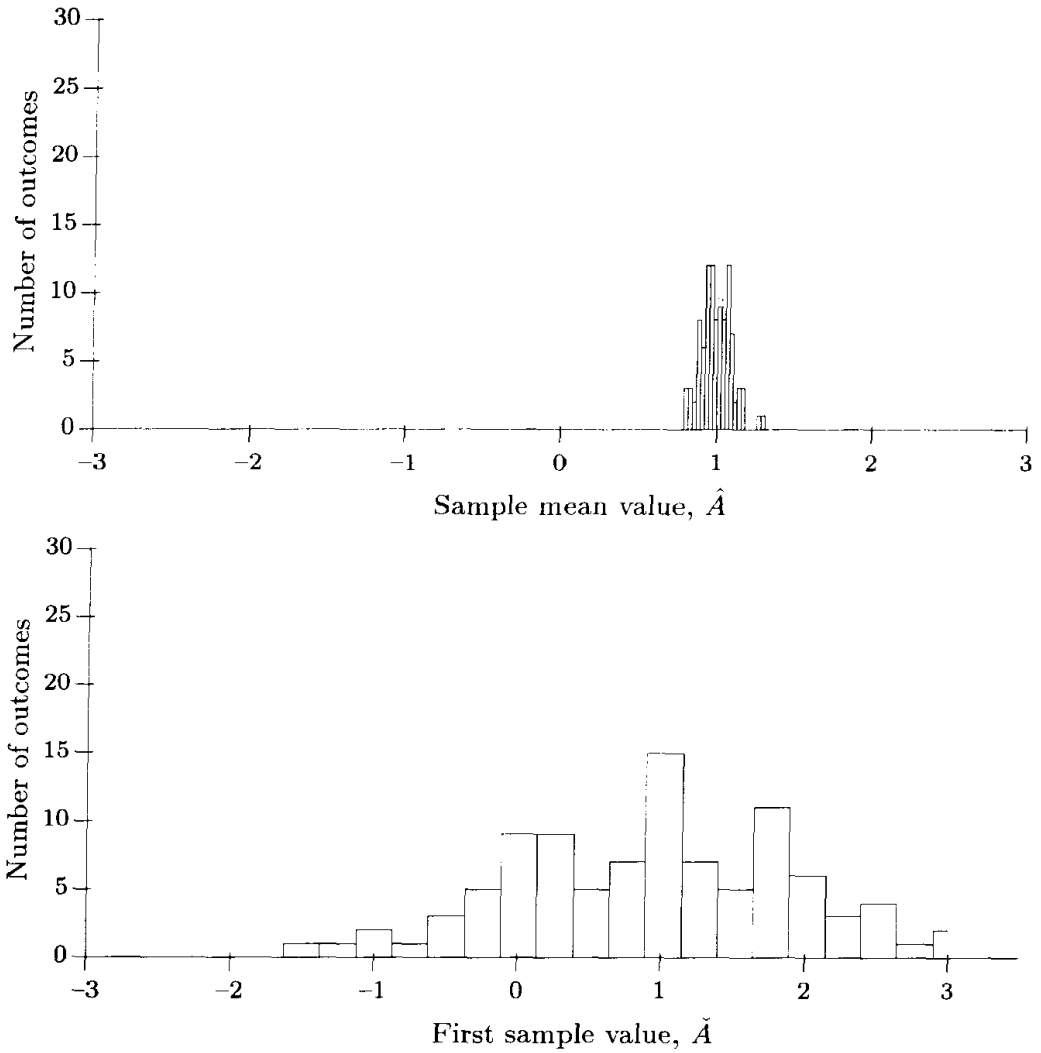


Figure 1.8 Histograms for sample mean and first sample estimator

$$\begin{aligned} &= \frac{1}{N^2}\sum_{n=0}^{N-1}\text{var}(x[n]) \\ &= \frac{1}{N^2}N\sigma^2 \\ &= \frac{\sigma^2}{N} \end{aligned}$$

since the $w[n]$'s are uncorrelated and thus

$$\begin{aligned} \text{var}(\check{A}) &= \text{var}(x[0]) \\ &= \sigma^2 \\ &> \text{var}(\hat{A}). \end{aligned}$$

Furthermore, if we could assume that $w[n]$ is Gaussian, we could also conclude that the probability of a given magnitude error is less for $\hat{A}$ than for $\check{A}$ (see Problem 2.7).

Several important points are illustrated by the previous example, which should always be kept in mind.

1. An estimator is a random variable. As such, its performance can only be completely described *statistically* or by its PDF.

2. The use of computer simulations for assessing estimation performance, although quite valuable for gaining insight and motivating conjectures, is never conclusive. At best, the true performance may be obtained to the desired degree of accuracy. At worst, for an insufficient number of experiments and/or errors in the simulation techniques employed, erroneous results may be obtained (see Appendix 7A for a further discussion of Monte Carlo computer techniques).

Another theme that we will repeatedly encounter is the tradeoff between performance and computational complexity. As in the previous example, even though $\hat{A}$ has better performance, it also requires more computation. We will see that optimal estimators can sometimes be difficult to implement, requiring a multidimensional optimization or integration. In these situations, alternative estimators that are suboptimal, but which can be implemented on a digital computer, may be preferred. For any particular application, the user must determine whether the loss in performance is offset by the reduced computational complexity of a suboptimal estimator.

1.4 Some Notes to the Reader

Our philosophy in presenting a theory of estimation is to provide the user with the main ideas necessary for determining *optimal* estimators. We have included results that we deem to be most useful in practice, omitting some important theoretical issues. The latter can be found in many books on statistical estimation theory which have been written from a more theoretical viewpoint [Cox and Hinkley 1974, Kendall and Stuart 1976–1979, Rao 1973, Zacks 1981]. As mentioned previously, our goal is to obtain an optimal estimator, and we resort to a suboptimal one if the former cannot be found or is not implementable. The sequence of chapters in this book follows this approach, so that optimal estimators are discussed first, followed by approximately optimal estimators, and finally suboptimal estimators. In Chapter 14 a "road map" for finding a good estimator is presented along with a summary of the various estimators and their properties. The reader may wish to read this chapter first to obtain an overview.

We have tried to maximize insight by including many examples and minimizing long mathematical expositions, although much of the tedious algebra and proofs have been included as appendices. The DC level in noise described earlier will serve as a standard example in introducing almost all the estimation approaches. It is hoped that in doing so the reader will be able to develop his or her own intuition by building upon previously assimilated concepts. Also, where possible, the scalar estimator is

presented first followed by the vector estimator. This approach reduces the tendency of vector/matrix algebra to obscure the main ideas. Finally, classical estimation is described first, followed by Bayesian estimation, again in the interest of not obscuring the main issues. The estimators obtained using the two approaches, although similar in appearance, are fundamentally different.

The mathematical notation for all common symbols is summarized in Appendix 2. The distinction between a continuous-time waveform and a discrete-time waveform or sequence is made through the symbolism $x(t)$ for continuous-time and $x[n]$ for discrete-time. Plots of $x[n]$, however, appear continuous in time, the points having been connected by straight lines for easier viewing. All vectors and matrices are boldface with all vectors being *column* vectors. All other symbolism is defined within the context of the discussion.

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Problems

1. In a radar system an estimator of round trip delay τ_0 has the PDF $\hat{\tau_0} \sim \mathcal{N}(\tau_0, \sigma_{\hat{\tau_0}}^2)$, where τ_0 is the true value. If the range is to be estimated, propose an estimator $\hat{R}$ and find its PDF. Next determine the standard deviation $\sigma_{\hat{\tau_0}}$ so that 99% of the time the range estimate will be within 100 m of the true value. Use $c = 3 \times 10^8$ m/s for the speed of electromagnetic propagation.

2. An unknown parameter θ influences the outcome of an experiment which is modeled by the random variable x. The PDF of x is

$$p(x;\theta) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(x-\theta)^2\right].$$

   A series of experiments is performed, and x is found to always be in the interval [97, 103]. As a result, the investigator concludes that θ must have been 100. Is this assertion correct?

3. Let $x = \theta + w$, where w is a random variable with PDF $p_w(w)$. If θ is a deterministic parameter, find the PDF of x in terms of p_w and denote it by $p(x;\theta)$. Next assume that θ is a random variable independent of w and find the conditional PDF $p(x|\theta)$. Finally, do not assume that θ and w are independent and determine $p(x|\theta)$. What can you say about $p(x;\theta)$ versus $p(x|\theta)$?

4. It is desired to estimate the value of a DC level A in WGN or

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

   where $w[n]$ is zero mean and uncorrelated, and each sample has variance $\sigma^2 = 1$. Consider the two estimators

$$\begin{aligned} \hat{A} &= \frac{1}{N}\sum_{n=0}^{N-1} x[n] \\ \check{A} &= \frac{1}{N+2}\left(2x[0] + \sum_{n=1}^{N-2} x[n] + 2x[N-1]\right). \end{aligned}$$

   Which one is better? Does it depend on the value of A?

5. For the same data set as in Problem 1.4 the following estimator is proposed:

$$\hat{A} = \begin{cases} x[0] & \frac{A^2}{\sigma^2} = A^2 > 1000 \\ \frac{1}{N}\sum_{n=0}^{N-1} x[n] & \frac{A^2}{\sigma^2} = A^2 \leq 1000. \end{cases}$$

   The rationale for this estimator is that for a high enough signal-to-noise ratio (SNR) or A^2/σ^2, we do not need to reduce the effect of noise by averaging and hence can avoid the added computation. Comment on this approach.